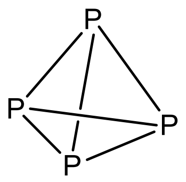
Section A (60 marks)

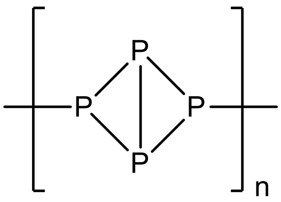
Answer **all** the questions in this section in the space provided.

1 Phosphorus and sulfur are elements in Period 3 of the Periodic Table and each can exist in various allotropic forms.

Phosphorus can exist as white phosphorus and red phosphorus. White phosphorus exists as a tetrahedron with a bond angle 60° while red phosphorus exists as a polymeric chain of regular tetrahedrons.



white phosphorus



red phosphorus

(a) The bond angle in red phosphorus is 80°C **[2]**

By considering the number of bond pairs and lone pairs, suggest why white phosphorus is less stable than the red phosphorus.

Phosphorus has 1 Lone pair and 3 bond pairs/ or the optimal bond angle should be 107°. The bond angle of 60° in white phosphorus, P_4 is smaller than the angle in red phosphorus. Hence, there will be ring strain in the bonding / strong electronic repulsion between the bond pairs.

(b) The boiling points of the two chlorides of phosphorous are shown in the table below.

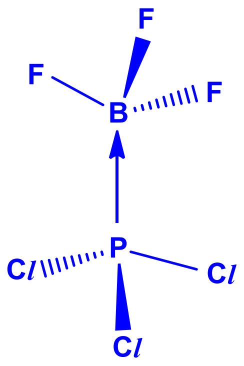
compound	boiling point / °C
PCl_3	76.1
PCl_5	166.8

(i) With reference to the structure and bonding, explain the difference in boiling points between the two chlorides of phosphorous. **[2]**

Both compounds are simple covalent compound. Due to the greater number of electrons in PCl_5, more energy is required to overcome the stronger instantaneous dipole–induced dipole interactions in PCl_5

compared to the permanent dipole–permanent dipole interactions in PCl_3

(ii) PCl_3 is able to react with BF_3. State the type of bond formed and draw a diagram to illustrate the shape of the product obtained. **[2]**



(c) A sample of pure sulfur which contains only ^{32}S and ^{33}S has a relative atomic mass of 32.08

(i) Define the term *relative atomic mass* of an element. **[1]**

The relative atomic mass of an element is defined as the ratio of the average mass of one atom of the element to $\frac{1}{12}$ the mass of a ^{12}C atom.

(ii) Calculate the percentage abundances of both ^{32}S and ^{33}S in this sample of sulfur. **[1]**

$32x + 33(1-x) = 32.08$

$x = 0.92$

% of ^{32}S = 92.0%

% of ^{33}S = 8.0%

(d) When sulfur is heated under pressure with chlorine, the major product is SCl_2. Under suitable conditions, SCl_2 hydrolyse in water to produce sulfur, SO_2 and HCl. During the reaction, 35 $kJmol^{-1}$ of energy is released.

(i) Constructing an equation for the hydrolysis of SCl_2. **[1]**

$2SCl_2 + 2H_2O \rightarrow S + SO_2 + 4HCl$

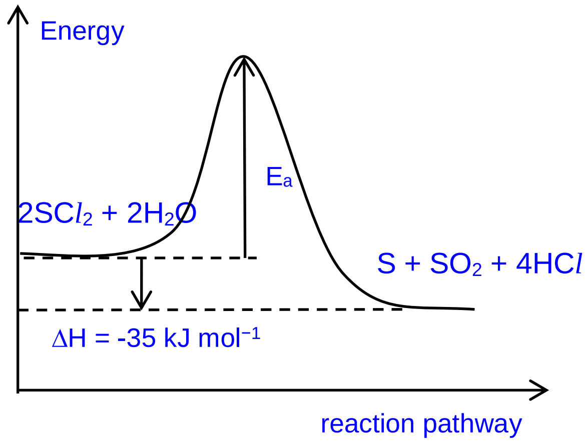
(ii) Hence, explain in terms of oxidation numbers, what happens to the sulfur in this reaction and state the type of reaction. **[2]**

$2SCl_2 + 2H_2O \rightarrow S + SO_2 + 4HCl$

Oxidation number of sulfur <u>increases from +2</u> (in SCl_2) <u>to +4</u> (in SO_2), and also <u>decreases from +2</u> (in SCl_2) <u>to 0</u> (in S).

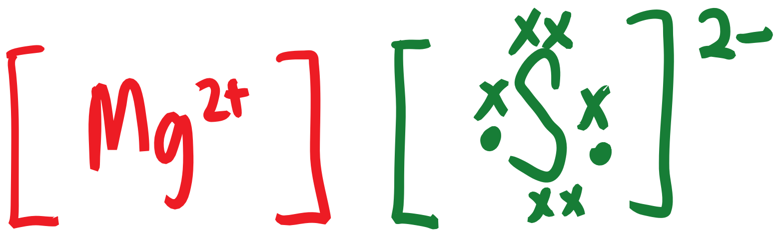
Type of reaction: Disproportionation

(iii) Draw an energy profile diagram for the reaction in **(e)**. Label the activation energy E_a, and the enthalpy of reaction ΔH. **[1]**



(e) Sulfur can react with magnesium to form magnesium sulfide, MgS.

(i) Draw a 'dot – and – cross' diagram for magnesium sulfide. **[1]**



(ii) Magnesium sulfide dissolves in water. **[2]**

Use your knowledge of the Periodic Table to predict a value for the pH of the solution formed. Explain how you arrived at your answer.

pH ≈ 9 Since sulfur and oxygen are in the same group, they would form similar anions, thus magnesium sulfide should behave like magnesium oxide to give an alkaline solution. [1 for correct pH and 1 explanation.]

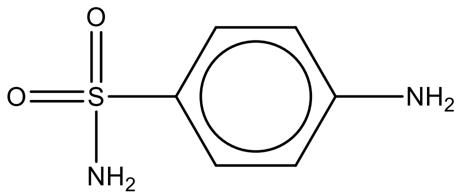
(iii) Explain why a sulfur atom has a smaller atomic radius than a magnesium atom but a sulfide ion has a larger ionic radius than a magnesium ion **[3]**

Sulfur has more protons than magnesium, hence stronger nuclear charge. Screening effect is not significantly larger for sulfur as the additional electrons are added to same valence shell. Hence, strong attraction between nucleus and valence electron for sulfur, leading to smaller atomic radius.

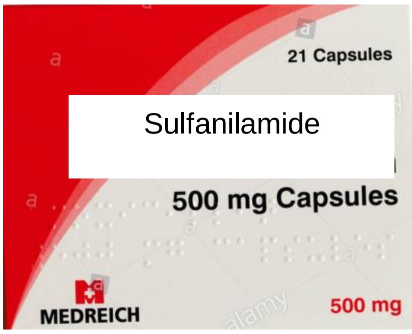
There is one extra shell of electrons in sulfide ion as compared to that of Mg ion. Hence, the valence electron in S^{2-} is further away from the nucleus and experience greater shielding effect, leading to weaker attraction of valence electrons. Hence, the ionic radius of sulfide ion is larger.

[Total: 18 marks]

2 Sulfanilamide, is an anti-bacterial drug, which had been used safely in tablet and powder form. However, in 1937, Elixir sulfanilamide, a medicine which was synthesised using diethylene glycol as a solvent, poisoned and killed more than 100 people because of acute kidney failure. It was found out later that it was the solvent that was poisonous.



sulfanilamide



The maximum daily dosage of sulfanilamide is 0.08 g per kg of patient. An average adult may be assumed to weigh about 70 kg.

(a) Is it safe for an infected person to consume 15 capsules in a single day? Justify with calculations. **[1]**

No

Maximum mass of daily dosage of sulfanilamide = 0.08 x 70 = 5.6 g

Maximum number of tablets = 5.6/0.5 =11.2 tablets. < 15

(b) **(i)** Assuming sulfanilamide is a monoacidic base, write an equation to show how sulfanilamide react with aqueous HC*l*. **[1]**

You may use RNH_2 to represent sulfanilamide.

$RNH_2 + H^+ \rightarrow RNH_3^+$

(ii) Hence, explain why sulfanilamide is soluble in acids but not in water. **[1]**

Sufanilamide is a base / will be protonated by acid. Hence, it will form stronger ion-dipole interactions with water as compared to hydrogen bonding.

(c) 0.15 g of an antibiotic power containing sulfanilamide was dissolved in 100 cm^3 of distilled water to form an aqueous solution.

A 25.0 cm^3 aliquot was transferred into a conical flask, in which 10.0 cm^3 of 0.0250 mol dm^{-3} $KBrO_3$ was added. About 10 g of solid KBr was then added.

BrO_3^- reacts with bromide according to the equation:

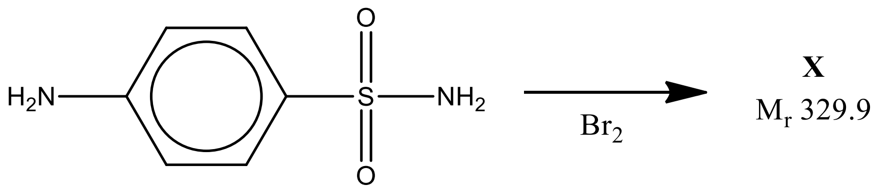
$$BrO_3^- + 5Br^- + 6H^+ \rightarrow 3Br_2 + 3H_2O$$

(i) Calculate the number of moles of Br_2 formed. **[1]**

Amount of Br_2 = 3 x (10.0 x 0.0250 / 1000) = 0.000750 mol

The bromine formed then reacts with the sulfanilamide to form **X**. In this reaction, the bromine atom substitutes the H in the benzene ring.

Note that the following equation is not balanced.



sulfanilamide
M_r 172.1

(ii) By comparing the M_r of sulfanilamide and **X**, deduce whether **X** is a mono-brominated or di-brominated compound. Show your working. **[1]**

X is a dibrominated compound.

The A_r of bromine is 79.9 to the nearest 1 dp.

Observing the 0.9 in the M_r of **X**, we can conclude there are two bromine atoms in **X**.

Or

329.9 + 2 – (172.1) = 159.8 = 2(79.9)

(iii) After 15 minutes, an excess of KI was added. **[1]**

$$Br_2 + 2I^- \rightarrow 2Br^- + I_2$$

The liberated iodine was then titrated with 15.00 cm^3 of 0.08500 mol dm^{-3} $Na_2S_2O_3$.

$$I_2 + 2S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2I^-$$

Calculate the number of moles of bromine, which reacted with the KI.

Amt of bromine = Amt of iodine = 1/2 x (15.00 x 0.0850 / 1000) = 6.375×10^{-4} mol = amount of bromine reacted with KI.

(iv) Using your answers to **(c)(i)** and **(c) (iii)**, calculate the number of moles of sulfanilamide which reacted with the bromine. **[1]**

Number of moles of bromine reacted with sulfanilamide

$= (7.50 \times 10^{-4}) - (6.375 \times 10^{-4}) = 1.125 \times 10^{-4}$ mol

(v) Assuming the mole ratio of bromine:sulfanilamide = 1:1, calculate the percentage by mass of sulfanilamide in the sample. **[2]**

Mass of sulfanilamide in sample = $1.13 \times 10^{-4} \times 100/25 \times 172.1 = 0.0777$ g

Percentage mass of sulfanilamide in sample = $\frac{0.0777\ g}{0.15g} \times 100\% = 51.8\%$

[Total: 9marks]

3 (a) The halogens, found in group 17 of the Periodic Table, and their compounds are useful laboratory reagents.

The table below shows some properties of the halogens and their compounds.

halogen	chlorine	bromine	iodine
standard enthalpy change of vapourisation ($\Delta H_v^\ominus$) of X_2 / kJ mol^{-1}	0	+15	+30
electronegativity	3.2	3.0	2.7
standard enthalpy change of reaction ($\Delta H_r^\ominus$) for thermal decomposition of HX / kJ mol^{-1}	+183	+103	+11

By quoting relevant data from the *Data Booklet*, explain the standard enthalpy change of reaction ($\Delta H_r^\ominus$) for thermal decomposition of the hydrides of the Group 17 elements. **[2]**

hydrogen halides	H−C*l*	H−Br	H−I
bond energy / kJ mol^{-1}	431	366	299

Order of thermal stability : HC*l* > HBr > HI

Down the group,

- Atomic radius increases, less effective overlap between atomic orbitals
- bond strength decreases
- less energy required to break H-X bond
- Thus lower ($\Delta H_r^\ominus$) for thermal decomposition of HX

(b) Astatine, At, is found below iodine in group 17 of the Periodic Table. Using your knowledge of the chemistry of group 17, predict the changes (if any) that you may observe when the following reagents are mixed.

If nothing is formed, write “no reaction” and where there is a reaction predicted, write an ionic equation with state symbols.

(i) aqueous bromine and aqueous sodium astatide [1]

Decolourisation of orange/yellow/reddish brown $Br_2(aq)$ and a black solid is formed.

$Br_2(aq) + 2At^-(aq) \rightarrow 2Br^-(aq) + At_2(s)$

(ii) astatine and aqueous sodium chloride [1]

no reaction

(c) When sodium iodide reacts with concentrated sulfuric acid, iodine is formed as one of the products. [1]

When sodium chloride reacts with concentrated sulfuric acid, chlorine is **not** formed as one of the product.

Suggest why when concentrated sulfuric acid is reacted with these halide salts, iodine is formed but chlorine is not.

Iodide is a <u>stronger reducing agent</u> than chloride as it is <u>more easily oxidised</u> than chloride.

[Total: 5 marks]

4 (a) The Haber-Weiss reaction generates highly reactive free radicals from H_2O_2. A free radical is formed when the O–O covalent bond in H_2O_2 breaks. Explain why the free radical is highly reactive. [1]

The free radical is highly reactive as it has a single/unpaired electron

(b) Hydrogen peroxide can undergo disproportionation reaction in the presence of the Fe^{2+} catalyst.

$$H_2O_2 \longrightarrow H_2O + ½O_2$$

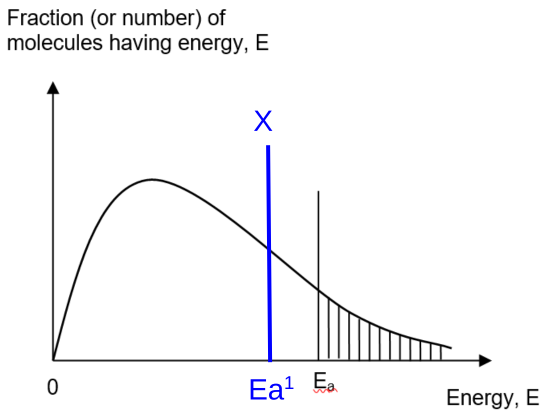
The uncatalysed reaction has an activation energy of +49 kJ mol^{-1}.

(i) Define the term activation energy. [1]

Activation energy is the <u>minimum</u> energy molecules need to possess for a reaction to occur.

(ii) Draw and label a Boltzmann distribution curve for the molecular energies of a sample of a H_2O_2 mixture. **[2]**

On your diagram, draw a line to represent the activation energy. Label this line E_a.



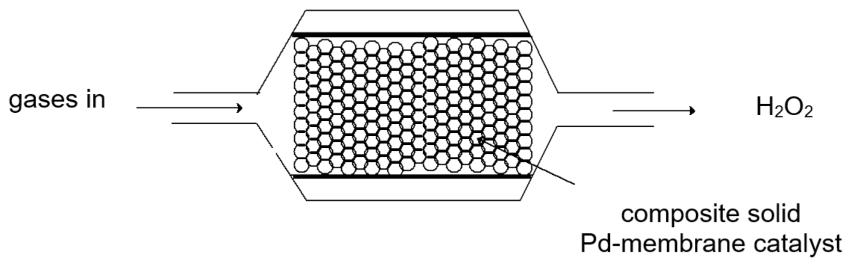
correct label of axis +

shape of curve starting from origin + label Ea

(iii) On the Boltzmann distribution curve in **b(ii)** show the change that would happen when a catalyst is added to the reaction mixture given that the temperature and pressure are kept constant. Label the change X. **[1]**

(c) H_2O_2 is in high demand in chemical industries as it is considered an environmentally-friendly oxidising agent.

In recent times, a novel composite solid palladium membrane catalyst for the direct oxidation of H_2 to H_2O_2 has been used.



(i) Suggest why it is environmentally-friendly to use H_2O_2 as an oxidising agent. **[1]**

The product of oxidation is water/oxygen which is clean/non-pollutants/environmentally friendly.

(ii) State what is meant by the term *heterogeneous* when it is applied to a Pd catalyst in the oxidation reaction of H_2 to H_2O_2. **[1]**

Heterogenous catalyst refers to catalyst not in the same phase as the reactants. In this case, Pd is a solid while the reactants (H2 and O2) are gases.

(iii) Outline, in three stages, the mode of action as Pd catalyses the oxidation reaction of H_2. **[3]**

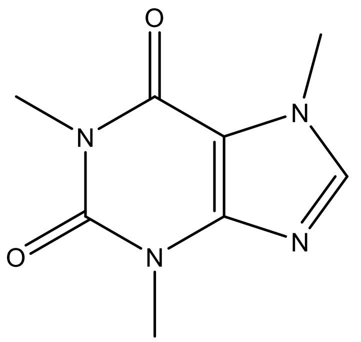
Stage 1: Pd provide a surface for the reactant molecules to be adsorbed onto it.

Stage 2: This weaken the bonds within the reactant molecules and they eventually break down.

Stage 3: The atoms then form new bonds to form the products and the products desorbs away from the catalyst for it to be reused again.

[Total: 10 marks]

5 Holly berries contain bitter compounds such as alkaloids, which make them mildly toxic if eaten. One alkaloid culprit is theobromine, which is also found in small amounts in chocolate.



theobromine

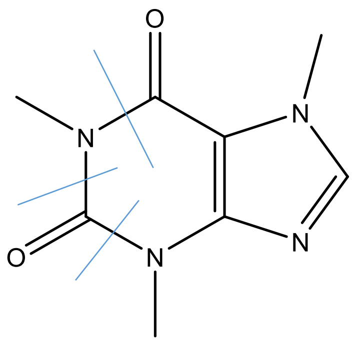
Figure 5.1

(a) **(i)** Deduce the molecular formula of theobromine. **[1]**

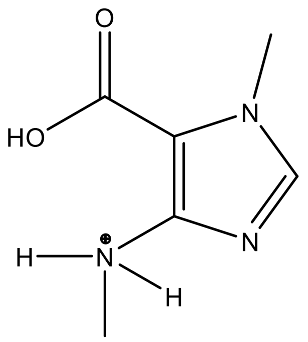
$C_8H_{10}N_4O_2$

(ii) Theobromine can be hydrolysed. **[2]**

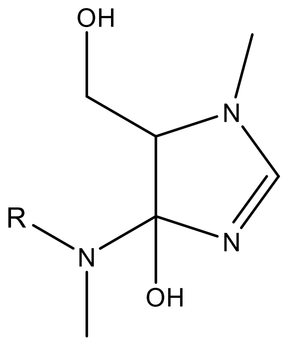
Draw lines on Figure 5.1 through all the bonds which are broken during the complete hydrolysis of theobromine.



(iii) Draw the structure of the acidic hydrolysis product with the biggest *Mr*. **[1]**



(b) Compound **X** is used in the synthesis of theobromine.



Compound **X**

Use the table of characteristic infra-red absorption frequencies in the *Data Booklet* to answer this question. **[1]**

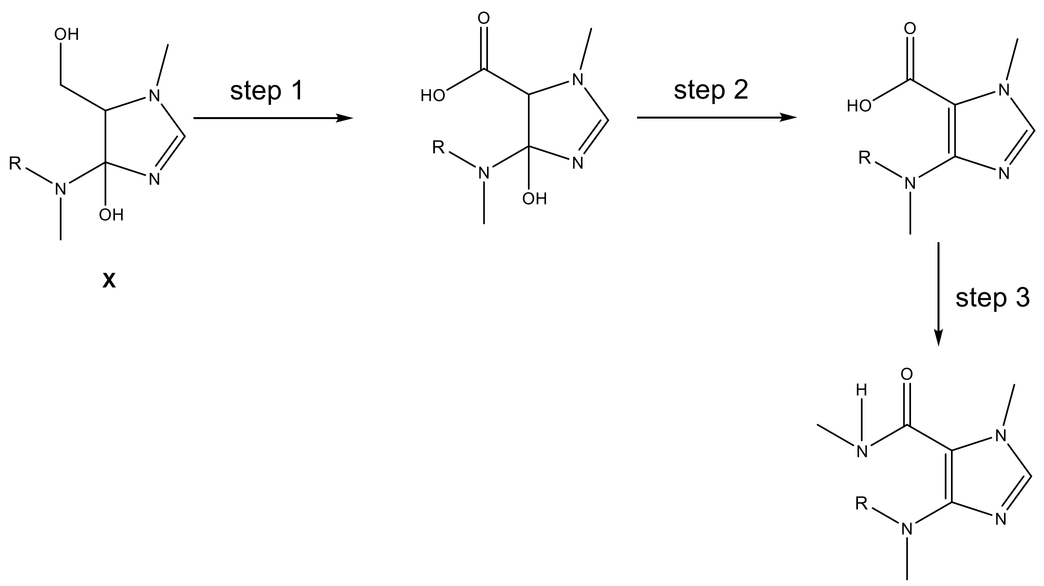
Use the table to identify an infra-red absorption range that will be shown by compound **X** but not by theobromine

.... cm^{-1}

970-1260 (C-O in alcohol)

3580-3650 (O-H in alcohol)

(c) The following method was proposed by a student to show part of the synthesis of theobromine from compound **X**.



(i) State the reagents and conditions for steps 1 to 3. **[3]**

Step 1:

………………………………………………………………….……………………

Step 2:

………………………………………………………………….……………………

Step 3:

………………………………………………………………….……………………

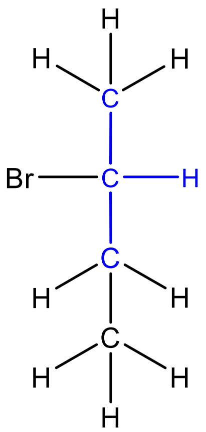
Step 1: $K_2Cr_2O_7$, H_2SO_4(aq), heat or $KMnO_4$,H_2SO_4(aq), heat

Step 2: excess conc H_2SO_4, heat

Step 3: CH_3NH_2, DCC

(d) There are four isomeric bromoalkanes with the formula C_4H_9Br. They can all form alkenes when undergoing elimination reactions. **[2]**

Name and draw the displayed formula of the isomer, which forms an alkene that exhibits *cis-trans* isomerism.



Name: 2- bromobutane

[Total: 10 marks]

6. Solid hydrated barium hydroxide, $Ba(OH)_2.8H_2O$ reacts with solid ammonium chloride at room temperature according to the following equation.

$$Ba(OH)_2.8H_2O(s) + 2NH_4Cl(s) \rightarrow 2NH_3(g) + 10H_2O(l) + BaCl_2(s)$$

	ΔH_f / $kJmol^{-1}$
$Ba(OH)_2.8H_2O(s)$	-3345
$NH_4Cl(s)$	-314
$NH_3(g)$	-46
$H_2O(l)$	-286
$BaCl_2(s)$	-859

Table 6.1

(a) Use the equation and the data given in **Table 6.1** to calculate the enthalpy change of reaction for the reaction. **[2]**

ΔH = [2(-46)+10(-286)+(-859)] – [(-3345)+2(-314)] [1]

= - 3811 + 3973

= +162 kJ mol^{-1} [1]

(b) **(i)** Define the term lattice energy **[1]**

Lattice energy is the enthalpy change when one mole of pure solid ionic compound is formed from its gaseous ions.

(ii) State how the magnitude of the lattice energy of $BaCl_2(s)$ compares to that of $NH_4Cl(s)$. Explain your answer. **[2]**

$|\text{Lattice Energy}| \propto \left|\frac{q^+q^-}{r_+ + r_-}\right|$

The anion is the same for both species.

Ba^{2+} has a higher charge [1]and a smaller size. hence the magnitude of the lattice energy of $BaCl_2(s)$ is greater[1] than that of $NH_4Cl(s)$.

(c) Barium hydroxide is a strong base.

Calculate the pH of 0.015mol dm^{-3} $Ba(OH)_2$. **[2]**

$[OH^-] = 2 \times 0.015 = 0.03 \text{ mol dm}^{-3}$ [1]

pH = -lg 0.03 = 1.52

pOH = 14 – 1.52 = 12.48 [1]

(d) *Use of the Data Booklet is relevant to this question.*

$^{137}_{56}Ba$ is able to undergo a process known as alpha(α)-decay in which a helium nucleus, $^{4}_{2}He$, and another species is emitted.

Complete the equation for the alpha(α)decay of $^{137}_{56}Ba$. **[1]**

$$^{137}_{56}Ba \longrightarrow \, ^{133}_{54}Xe + \, ^{4}_{2}He$$

Section B

Answer **one** question from this section in the spaces provided.

7. (a) *t*-butyl chloride reacts with alcoholic sodium methoxide, $CH_3O^-Na^+$, to form 2-methylpropene.

$$\underset{t\text{-butyl chloride}}{(CH_3)_3CCl} + CH_3O^- \longrightarrow \underset{\text{2-methylpropene}}{(CH_3)_2C{=}CH_2} + Cl^- + CH_3OH$$

The kinetics of this reaction was studied and the following results were obtained.

experiment	$[(CH_3)_3CCl]$ /mol dm^{-3}	$[CH_3O^-]$ /mol dm^{-3}	Initial rate /mol dm^{-3} min^{-1}
1	0.0500	0.0400	4.00×10^{-6}
2	0.0300	0.0400	2.40×10^{-6}
3	0.0200	0.0300	1.60×10^{-6}

Table 7.1

(i) State the type of reaction when *t*-butyl chloride reacts with alcoholic sodium methoxide. **[1]**

Elimination

(ii) Using the data in **Table 7.1**, deduce the order of reaction with respect to $[(CH_3)_3CCl]$ and $[CH_3O^-]$. **[2]**

Comparing expt 1 and 2, when $[(CH_3)_3CCl]$ x 3/5, rate x 3/5. Hence order of reaction with respect to $[(CH_3)_3CCl]$ is 1.

Comparing expt 1 and 3, $[(CH_3)_3CCl]$ x 2/5, rate x 2/5. Since order of reaction with respect to $[(CH_3)_3CCl]$ is 1, the change in $[CH_3O^-]$ has no effect on the rate Hence order of reaction with respect to $[CH_3O^-]$ is 0.

OR

$$\frac{(0.02)(0.03)^x}{(0.05)(0.04)^x} = \frac{1.6 \times 10^{-6}}{4 \times 10^{-6}}$$

$$(\frac{2}{5})(\frac{3}{4})^x = \frac{2}{5}$$

x = 0

(iii) Using Experiment 1, calculate a value for the rate constant. Include units in your answer. **[2]**

Rate = $k[(CH_3)_3CCl]$

Using expt 1,

$4.00 \times 10^{-6} = k(0.0500)$

$k = 8.00 \times 10^{-5}$

units = min^{-1}

(iv) Suggest a chemical test to distinguish *t*-butyl chloride and 2-methylpropene. State any observations you would make with each compound. **[2]**

Aq Bromine

Orange aq bromine decolourise for 2-methylpropene. Orange aq bromine remains for t-butyl chloride.

OR

Ethanolic $AgNO_3$, heat

t-butylchloride forms white ppt but no white ppt for 2-methylpropene.

(b) Aqueous sodium vanadate(V) contains VO_3^- ions. These vanadate(V) ions exist in equilibrium with vanadate(IV) ions as shown by the expression below.

$$VO_3^-(aq) + 4H^+(aq) + e \rightleftharpoons VO^{2+}(aq) + 2H_2O(l)$$

yellow   blue

(i) State the full electronic configuration of the vanadium ion in VO_3^-. **[1]**

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

(ii) Write an expression for the equilibrium constant, K_c, for this reaction. **[1]**

$Kc = \frac{[VO^{2+}]}{[VO_3-][H+]^4}$

(iii) A solution is prepared in which the initial concentration of VO_3^- ions and H^+ ions are both 1.00 moldm^{-3}. After equilibrium is reached, the concentration of VO^{2+} ions is 0.125 moldm^{-3}.

Calculate the value of K_c and state its units. **[3]**

	$VO_3^-(aq)$	$+ 4H^+(aq) + e \rightleftharpoons$	$VO^{2+}(aq) + 2H_2O(l)$
Initial conc	1.00	1.00	0
Change in conc	-0.125	-0.5	+0.125
Eqm conc	0.875	0.5	0.125

[1] for change in concentration

$Kc = \frac{0.125}{0.874 \times 0.5^4}$ = 2.29 mol^{-4} dm^{12}

(iv) If water is added to the vessel containing VO_3^-(aq) ions and VO^{2+}(aq) ions in equilibrium, the green solution becomes more yellow.

Explain the observation. **[2]**

<u>More concentration terms/ions</u> on the left hand side of the equation.

The ions on the left hand side are <u>diluted to a greater extent</u>.

<u>Equilibrium position will shift to the left to increase the concentration of the ions on the left hand side</u>. Hence green solution will become more yellow.

(c) In 2021, Singapore built one of the world's biggest floating solar farms. 122 000 solar panels on the surface of Tengah Reservoir will power PUB's water treatment plants. The panels, made of glass and metal, are supported by floats made of food-grade certified high density polyethylene, HDPE, to ensure that water quality is not compromised. They are also UV-resistant and can withstand intense sunlight.

https://www.straitstimes.com/multimedia/graphics/2021/05/singapore-largest-solar-farm-water/index.html?shell

(i) Describe the structure of HDPE. **[2]**

Long and unbranched/linear chains formed from ethene monomers. Instantenous dipole induced dipole between polymeric covalent chains.

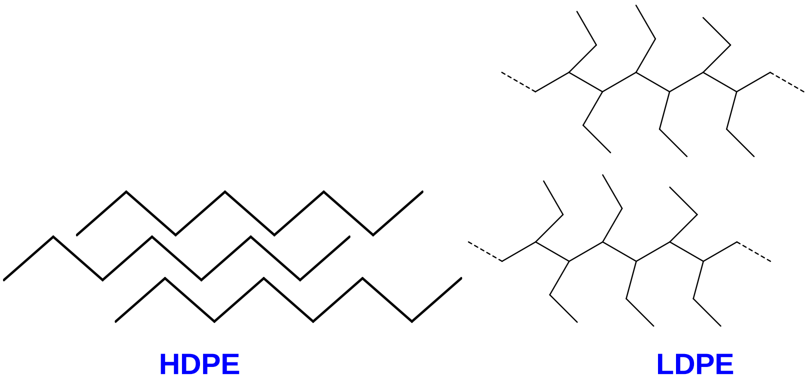
(ii) A material is food grade certified if the material is safe to come into direct contact with food products. Suggest why HDPE is food grade certified. **[1]**

Covalent structure with strong C-C and C-H covalent bonds hence inert and would not break down easily/Absence of ester or amide linkages

(iii) Suggest why HDPE is used over LDPE when used as a float to support the heavy weight of the solar panel **[1]**

HDPE is harder and stiffer/rigid and strong and can withstand the mass of the heavy solar panels

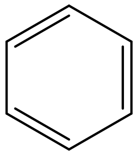
(iv) Draw labelled diagrams to illustrate the difference between HDPE and low density polyethylene, LDPE. **[2]**



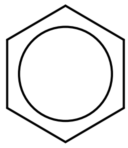
[Total: 20 marks]

8. In the past, several different structures have been suggested for benzene.

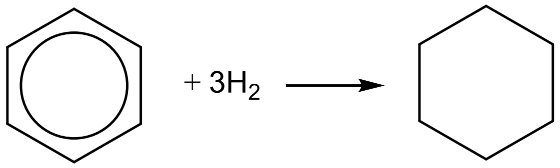
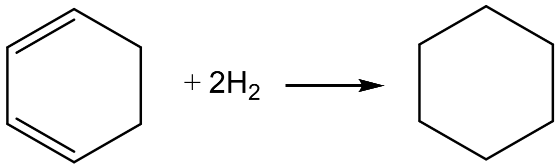
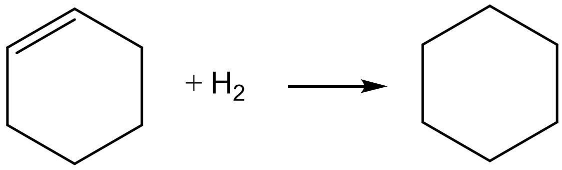
One of the suggested structures, which has been proved to be incorrect, was a compound containing three fixed double bonds, cyclohexa-1,3,5-triene.



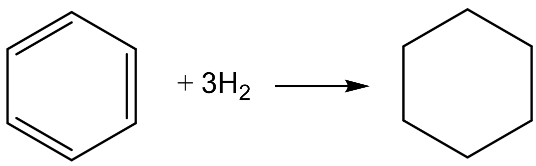
The currently accepted structure for benzene can be represented as follows.



Benzene and the unsaturated cyclic hydrocarbons can be reduced to cyclohexane as shown in the following equations.

	$\Delta H^{\ominus} = -208$ kJmol^{-1}
	$\Delta H^{\ominus} = -232$ kJmol^{-1}
	$\Delta H^{\ominus} = -120$ kJmol^{-1}

(a) **(i)** Estimate the value of the standard enthalpy change of the reaction below when cyclohexa-1,3,5-triene is reduced to cyclohexane. **[1]**



–360 kJ mol^{-1}

(ii) Explain your answer to **(a)(i).** **[1]**

When cyclohexa–1,3,5–triene containing three mole of alkene functional groups undergoes reduction with three moles of $H_2(g)$, standard enthalpy change of reaction was 3(–120) = –360 kJ mol^{-1}.

(iii) Explain why benzene does not exist as the cyclohexa-1,3,5-triene, using the data in the question and your answer to **(a)(i)** **[2]**

The enthalpy change of reaction for the reduction of benzene to cyclohexane is –208 kJ mol^{-1}. There is a difference of –152 kJ mol^{-1} in the enthalpy change of reaction compared to the answer in (b)(i). This indicates a difference in the energetic stability/resonance stability of the structure of benzene versus cyclohexa–1,3,5–triene .

(iv) Name the other reagent needed for the reduction of cyclohexa-1,3,5-triene **[1]**

Ni catalyst

(b) **(i)** Describe the bonding in benzene in terms of orbital overlap. You may draw a diagram to illustrate your answer. **[3]**

Two of the orbitals of each C overlap head-on with the orbitals of two adjacent carbon atoms to form two C–C sigma bonds. Another orbital from C overlaps head-on with the s orbital of H atom to form a C–H sigma bond.

The p orbital of carbon overlaps side-on both above and below the ring of carbon atoms, with the (unhybridised) p orbitals of the 2 neighbouring carbon atoms, forming a ring of π (pi) electrons. These π electrons are delocalized.

(ii) Write the equation to represent the standard enthalpy change of combustion of benzene(*l*). **[1]**

$C_6H_6(l) + 15/2O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l)$

(iii) Use appropriate bond energy values from the *Data* Booklet to calculate a value for the enthalpy change of combustion for benzene. **[2]**

$\Delta H_r = \sum BE\text{ (reactants)} - \sum BE\text{(products)}$

$= 6(+520) + 6(+410) + 15/2(+496) – 6(460) – 12(805)$

$= -3120\ kJmol^{-1}$

Aqueous silver nitrate can be used to distinguish between bromobenzene and bromoethane.

(iii) With the aid of suitable equations, describe how silver nitrate can be used to distinguish between bromobenzene and bromoethane. State any observations. **[2]**

$$CH_3CH_2Br + OH^- \xrightarrow{\text{heat}} CH_3CH_2OH + Br^-$$

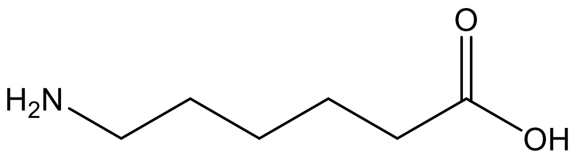
$$Ag^+ + Br^- \rightarrow \underset{\text{Cream ppt}}{AgBr}$$

(Iv) Explain why the substitution reaction with bromoethane occurs at a faster rate than the reaction with chloroethane. **[2]**

Bromine is a <u>bigger atom</u> hence it has a <u>less effective orbital overlap</u> with carbon. The <u>C-Br bond is weaker</u> and <u>easier to break</u> for substitution reaction.

(c) Many reusable face mask are made from nylon. Nylon is an example of a thermoplastic polymer.

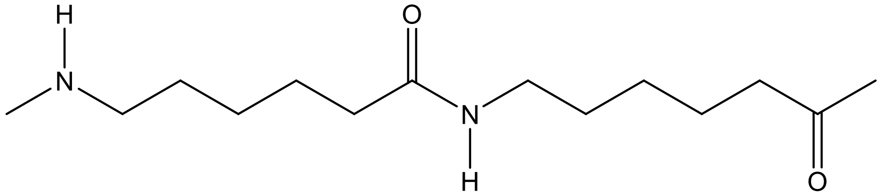
The monomer from which the polymer Nylon 6 is produced is showed below.



(i) Explain what is meant by the term thermoplastic **[1]**

Linear molecules linked together to form long linear chains with **no cross links**

(ii) Draw the structure of the Nylon 6, showing two repeat units. State the type of polymerisation involved. **[2]**



Condensation polymerisation

A typical washing instructions for such reusable face mask would read

"Machine wash cold, gentle cycle, do not bleach"

(iii) Suggest why the face mask should be washed in cold water during washing **[1]**

Hot water can cause hydrolysis of the amide linkage.

(iv) With reference to the structure of nylon 6, explain why a gentle cycle should be used when washing the face mask. **[1]**

When force is applied, the polymer is distorted and enough energy to break the intermolecular forces of attraction/hydrogen bonding/instantaneous dipole induced dipole interaction/covalent between the polymer chains

[Total: 20 marks]